

Riley Beenders

CONTROLS AUTOMATION | MOTION CONTROL | SOFTWARE-INTEGRATED SYSTEMS

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PROFESSIONAL SUMMARY

Controls and electromechanical R&D leader with 10+ years of multidisciplinary engineering experience, including 6+ years delivering large-scale automation, motion-control, and software-integrated machine systems. Leads control-system architecture, detailed design, safety and software risk analysis, integration, performance validation, troubleshooting, tuning, technical documentation, and commercialization.

CORE EXPERTISE

Controls & Motion: Control-system architecture and detailed design; industrial-servo and linear-motor systems; Allen-Bradley PLCs; CAN bus; Modbus sensor integration; performance analysis and validation; tuning and calibration

Software & Integration: LabVIEW; MATLAB, Simulink, and Simscape; Python; Visual Studio Code (IDE); Git/GitHub; software/hardware integration; debugging

Safety & Delivery: Safety, operational, and software risk analysis; system-level functional validation; fault diagnosis; technical writing and documentation; electrical schematics and BOM control; operator training; cross-functional leadership

PROFESSIONAL EXPERIENCE

Proteor / Proteor Printing Solutions | Coplay, PA

January 2025 - Present

Lead, Research & Development Engineer

- Lead the East Coast engineering team and collaborate across Proteor's global portfolio, owning controls-intensive products from architecture and detailed design through integration, validation, commercialization, and support.
- Led mechanical, electrical, motion-control, and software design of ICARUS-Lite, including 120/240 VAC operation; documented safety, operational, and software risk analyses across ICARUS-family products and ProteorPrint, with mitigations applicable to EU and U.S. standards.
- Converted POSEIDON and ICARUS from servo-driven ballscrews to a standardized linear-motor architecture; validated a 30 kg moving mass at speeds up to 4 m/s.
- Developed automated LabVIEW/Modbus metrology that turns nine readings from multiple laser sensors into live tolerance charts and PDF reports; led machine integration for an operator-facing remote operation and diagnostics interface.

Filament Innovations | Whitehall, PA

May 2020 - January 2025

Chief Technology Officer (Jan 2023 - Jan 2025) | Lead Electro-Mechanical Engineer (Jun 2022 - Jan 2023) | Industrial Engineer (Apr 2021 - Jun 2022) | Engineering Technician (May 2020 - Apr 2021)

- Directed product development and technology strategy while remaining hands-on across mechanical, electrical, firmware, software, and manufacturing systems.
- Designed and launched Gen3 POSEIDON with a 1 m³ hybrid-extrusion platform, industrial-servo XY motion, braked NEMA 34 Z drives for power-failure safety, and a custom gantry.
- Integrated and debugged a modular multi-board CAN-bus control architecture with independently testable electronics enclosures, simplifying wiring and enabling subsystem validation before installation.
- Delivered large automated systems end to end through CAD, BOMs, sourcing, fabrication, assembly, firmware tuning, calibration, operator training, and post-delivery support.

ADDITIONAL RELEVANT EXPERIENCE

Blue Mountain Ski Resort - Ski Lift Operator | 2019 - 2021 | Performed motor/control safety checks on passenger lifts and supported maintenance during breakdowns while assisting thousands of guests.

RJB.Engineering - Owner & Engineering Consultant | 2015 - Present | Deliver project-based engineering support for automotive clients and operate an online store for 3D-printing hardware accessories.

Northampton Area School District - Computer/Network Technician | 2016 - 2019 | Automated 2,500+ device deployments with an Arduino-based setup tool and deployed 750+ wireless access points.

EDUCATION & PROFESSIONAL DEVELOPMENT

Penn State University | B.S., Electro-Mechanical Engineering

December 2024

Relevant development: MathWorks MATLAB, Simulink, and Simscape Onramps (2026); Lean Six Sigma Yellow Belt (2025); Penn State Certificate in Engineering Design and Tools (2023)